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CO1 – AT2
MLA 0201 – Fundamentals of Machine Learning

Annexure A

Concept Mapping

1. Construct a concept map illustrating the complete Machine Learning workflow for predicting house prices using suitable ML algorithm.
2. Construct a concept map demonstrating how Classification algorithms are used in Medical Disease Prediction (e.g., Diabetes or Heart Disease Prediction).
3. Construct a concept map showing how Machine Learning is applied in Credit Card Fraud Detection.
4. Construct a concept map explaining Student Performance Prediction using the Multiple Linear Regression algorithm.
5. Construct a concept map showing how Decision Tree Classification is used for Loan Approval Predict

1. MACHINE LEARNING WORKFLOW FOR HOUSE PRICE PREDICTION

Aim

To demonstrate the complete Machine Learning workflow for predicting house prices using the Multiple Linear Regression algorithm.

Introduction

House Price Prediction is one of the most widely used applications of Machine Learning. Its main objective is to estimate the selling price of a house based on various factors such as area, number of bedrooms, number of bathrooms, location, age of the house, and other important property features. Since the predicted output is a continuous numerical value, Multiple Linear Regression is considered one of the most suitable algorithms for this prediction task.

Concept Explanation

The Machine Learning workflow begins with collecting housing data from reliable sources such as Kaggle datasets, government records, or real estate websites. After data collection, the dataset is preprocessed by handling missing values, removing duplicate entries, correcting inconsistent data, and converting categorical attributes into numerical values using suitable encoding techniques. Next, important features that have a significant impact on house prices are selected for model training. The prepared dataset is then divided into training and testing datasets. The Multiple Linear Regression algorithm is trained using the training data to learn the relationship between the input features and house prices. Once the model is trained, it predicts the prices of new houses based on their features. Finally, the model performance is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score to determine the prediction accuracy and overall model performance.

Algorithm Used:

Multiple Linear Regression

Applications

- House price prediction
- Real estate price estimation
- Property valuation
- Investment planning
- Market trend analysis

Conclusion

Machine Learning makes house price prediction faster and more accurate by learning patterns from historical housing data. Multiple Linear Regression identifies the relationship between different property features and their selling prices, enabling reliable predictions. This approach supports buyers, sellers, investors, and real estate companies in making better decisions based on data-driven insights.

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2. MEDICAL DISEASE PREDICTION USING CLASSIFICATION ALGORITHMS

Aim

To illustrate the complete Machine Learning workflow for predicting medical diseases using Classification algorithms.

Introduction

Medical Disease Prediction is one of the most important applications of Machine Learning in the healthcare field. The main objective is to predict whether a patient has a particular disease, such as diabetes or heart disease, based on medical attributes like age, blood pressure, glucose level, cholesterol, BMI, and other health-related factors. Since the output belongs to predefined categories (disease or no disease), Classification algorithms are the most suitable choice for this task.

Concept Explanation

The Machine Learning workflow begins with collecting patient health records from reliable sources such as hospitals, healthcare organizations, or publicly available datasets. The collected data is then preprocessed by handling missing values, removing duplicate records, correcting inconsistent data, and converting categorical attributes into numerical values whenever required. Important medical features that influence disease prediction are selected before dividing the dataset into training and testing sets. A suitable Classification algorithm, such as Decision Tree, Random Forest, Logistic Regression, or Support Vector Machine, is trained using the training data. After training, the model predicts whether a patient is likely to have the disease based on the given medical information. Finally, the model performance is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix to measure the effectiveness of the prediction.

Algorithm Used:

Classification Algorithms (Decision Tree, Random Forest, Logistic Regression, Support Vector Machine)

Applications

- Early disease detection
- Diabetes prediction
- Heart disease prediction
- Clinical decision support
- Healthcare risk assessment

Conclusion

Machine Learning improves disease prediction by identifying patterns from patient health records and medical data. Classification algorithms provide reliable predictions that help doctors detect diseases at an early stage, support accurate diagnosis, and improve treatment planning. This approach contributes to better healthcare services and informed medical decision-making.

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3. CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

Aim:

To demonstrate the complete Machine Learning workflow for detecting fraudulent credit card transactions using a Classification algorithm.

Introduction:

Credit Card Fraud Detection is one of the most significant applications of Machine Learning in the banking and financial sector. Its main objective is to identify whether a credit card transaction is genuine or fraudulent based on various transaction-related features such as transaction amount, transaction time, location, merchant details, spending behavior, and other important financial attributes. Since the predicted output is divided into two categories (Fraud or Not Fraud), Classification algorithms are considered the most suitable choice for this prediction task.

Concept Explanation:

The Machine Learning workflow begins with collecting credit card transaction data from reliable sources such as financial institutions, banking systems, or publicly available datasets. After data collection, the dataset is preprocessed by handling missing values, removing duplicate records, correcting inconsistent data, and converting categorical attributes into numerical values using suitable encoding techniques. The data is then balanced if required to handle the imbalance between fraudulent and genuine transactions. Important features that influence fraud detection are selected before dividing the dataset into training and testing datasets. A suitable Classification algorithm, such as Random Forest, Decision Tree, Logistic Regression, or Support Vector Machine, is trained using the training data to learn the transaction patterns. Once the model is trained, it predicts whether a new transaction is fraudulent or legitimate. Finally, the model performance is evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix to determine the detection accuracy and overall model performance.

Algorithm Used

Classification Algorithms (Random Forest, Decision Tree, Logistic Regression, Support Vector Machine)

Applications

- Credit card fraud detection
- Online payment security
- Banking transaction monitoring
- Financial risk management
- Digital fraud prevention

Conclusion

Machine Learning improves credit card fraud detection by identifying suspicious transaction patterns from historical financial data. Classification algorithms accurately distinguish between genuine and fraudulent transactions, helping banks and financial institutions reduce financial losses, enhance customer security, and ensure safe digital payment systems.

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4. STUDENT PERFORMANCE PREDICTION USING MULTIPLE LINEAR REGRESSION

Aim
To demonstrate the complete Machine Learning workflow for predicting student academic performance using the Multiple Linear Regression algorithm.

Introduction
Student Performance Prediction is one of the most useful applications of Machine Learning in the education sector. Its main objective is to predict a student's academic performance based on various factors such as study hours, attendance, previous examination marks, assignment scores, participation in class activities, and other important educational attributes. Since the predicted output is a continuous numerical value, Multiple Linear Regression is considered one of the most suitable algorithms for this prediction task.

Concept Explanation
The Machine Learning workflow begins with collecting student academic data from reliable sources such as schools, colleges, educational institutions, or publicly available datasets. After data collection, the dataset is preprocessed by handling missing values, removing duplicate records, correcting inconsistent data, and converting categorical attributes into numerical values using suitable encoding techniques. Next, important features that have a significant impact on student performance are selected for model training. The prepared dataset is then divided into training and testing datasets. The Multiple Linear Regression algorithm is trained using the training data to learn the relationship between the input features and student performance. Once the model is trained, it predicts the academic performance or final marks of students based on their learning-related features. Finally, the model performance is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score to determine the prediction accuracy and overall model performance.

Algorithm Used
Multiple Linear Regression
Applications

- Student performance prediction
- Academic progress monitoring
- Early identification of weak students
- Personalized learning support
- Educational decision-making

Conclusion
Machine Learning improves student performance prediction by identifying patterns from historical academic data. Multiple Linear Regression effectively determines the relationship between different educational factors and student outcomes, enabling accurate predictions. This approach helps teachers, students, and educational institutions make better decisions to improve learning performance and academic success.

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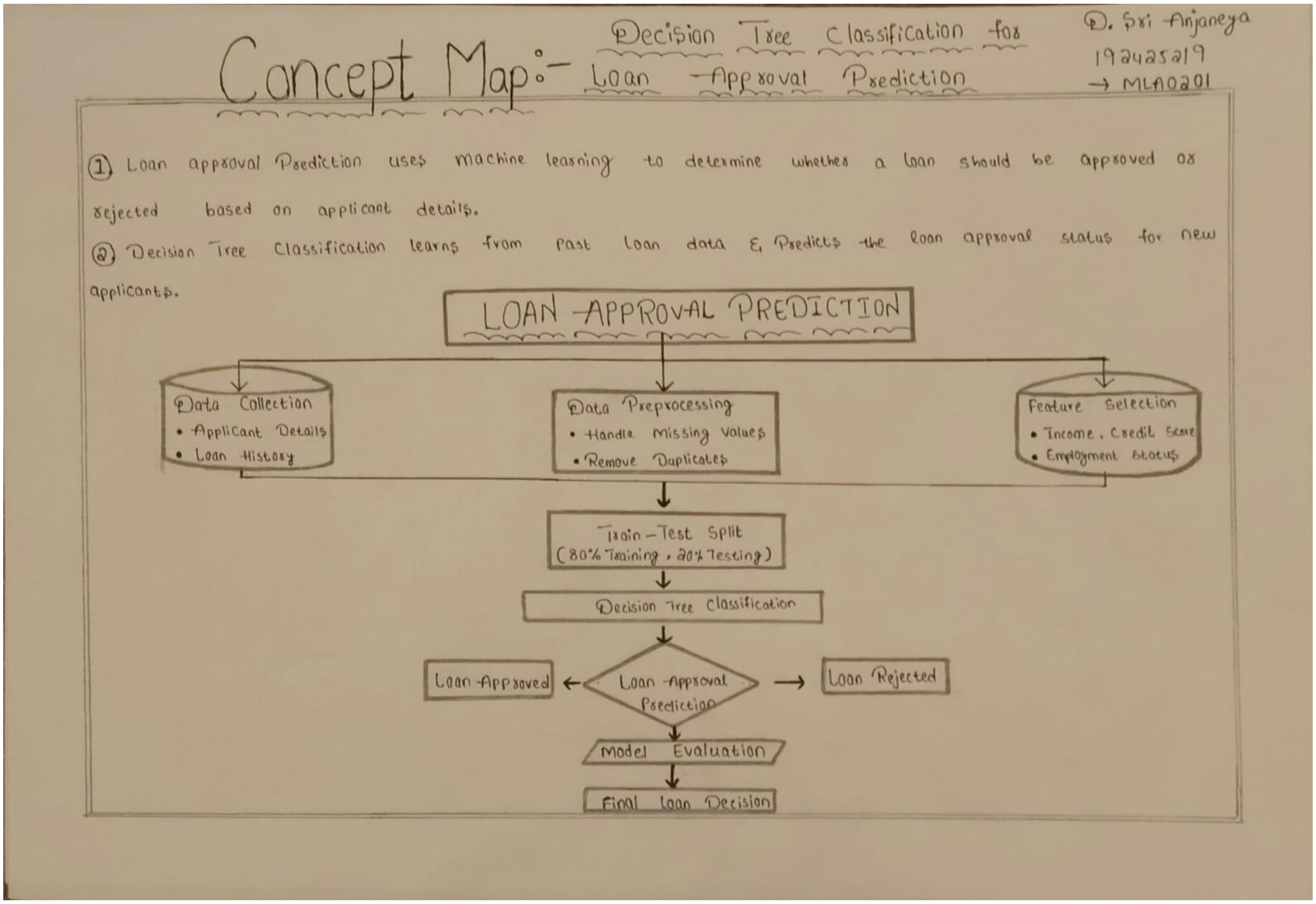
5. LOAN APPROVAL PREDICTION USING DECISION TREE CLASSIFICATION



Loan Approval Prediction using Decision Tree Classification is an important Machine Learning application used in the banking and financial sector. The main objective is to predict whether a loan application should be approved or rejected based on factors such as applicant income, credit score, loan amount, employment status, and repayment history. The Decision Tree Classification algorithm analyzes these features and creates decision rules to classify applicants into approved or rejected categories. This approach helps financial institutions make faster, accurate, and data-driven loan approval decisions while reducing risks.

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RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are wellstructured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

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